# Patient preferences for kidney failure treatments

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**This review considers what is important to people when making decisions about their kidney failure care. We begin by introducing the concept of preferences, and how preference-sensitive decision-making is central to person-centered care. Next, we explore the preferences that patients have in relation to kidney failure care and the attributes of available treatments. We examine how decisions are made about kidney failure treatments and consider whether preference-sensitive care is widespread. Finally, we make recommendations regarding how kidney services could be reshaped to ensure that people's preferences guide the health care they receive. We conclude that people's preferences can guide decisions between treatments for kidney failure only if clinicians and systems facilitate this. The best treatment for a person is the one that fits best with what is important to them. The *science* of nephrology involves optimal delivery of treatments that prolong life and preserve and replace organ function. The *art* lies in shaping this care to the person.**

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## Editor's Note

In this mini review, Hole *et al.* provide a comprehensive insight into decisions that frame preferences for kidney failure treatments. In doing so, they describe the *preference-sensitive* approach that complements the personalized care that is fundamental to optimal management of patients with advanced kidney disease. They highlight the intricate balance between the patient and the surrounding environment, including the clinicians, family members, and health care systems, which eventually leads to the most optimal shared decision-making for the benefit of the patient. The review offers a list of selected publications in table format, highlighting patient preferences across kidney failure therapies along with corresponding text that explains how these preferences lead to kidney failure treatment decisions. Given the potential deficiencies in preference-sensitive kidney care, the review is supplemented by recommendations as to how clinicians and services can reframe the culture and purpose of kidney care to be more about person-centeredness. Find the Management of Patients with Advanced Chronic Kidney Disease series at https://www.kidney-international.org/content/management-patients-with-ackd.

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recommendations regarding how kidney services could be reshaped to ensure that people's preferences guide the health care they receive.

## Preference-sensitive care

Person-centered care—high-quality, personalized health care that integrates people's values and beliefs—is desirable to patients and their families and has been incorporated into clinical practice guidelines.[1] Person-centered care seeks to align treatment decisions with patients' preferences. Commonly, the term *preference-sensitive* is restricted to treatment decisions with no clear *best* option.[2] However, as we will explore, the *best* treatment for an individual with kidney failure is determined by *their* preferences. This means person-centered services must consistently respond to what is important to patients.

A person's preference for something, including their treatment options, quantifies how desirable they consider that thing to be.[3] Preferences are subjective and personal, can be positive or negative, and can flux with time, setting, and context.[3] One person's preferences may seem bizarre to another, or even to themselves later on. People's choices reflect a *weighing up* of their preferences. When appraising goods or services, people consider the characteristics, or *attributes*, of alternatives. Attributes for which people have strong preferences exert greater *weight* on their decisions. When informed people are free to choose, their decisions can be used to infer what attributes were important to them. Specialist health economic surveys, including discrete choice experiments and best-worst scaling exercises, can be used to estimate how much of one attribute respondents are willing to relinquish to get more of another. Throughout this mini review, we have highlighted where such *choice experiments* have examined decision-making between kidney failure treatments.

Meanwhile, individuals' preferences are rarely the sole determinant of the health care they receive. Clinicians determine available treatments, and societal norms influence preferences. Decisions are often made at the level of the family rather than the individual. The decisions people make depend on the setting, the quality of the information they access, who they receive it from, and their emotional and cognitive state.[4] Families facing life-limiting illness confront fear, grief, guilt, distress, and anger—which can overwhelm deliberative *slow thinking* and lead to intuitive *fast reasoning*.[5] Patients are often asked to make decisions about health states and treatments of which they have no prior knowledge or experience while extremely unwell.[6] Preferences may change, especially with age or declining health.[7]

Shared decision-making aims to facilitate patient and health care professionals to work together to plan care that is informed by professional expertise and consistent with patients' preferences.[2] Approaches typically involve dialog regarding choice and options, elicitation of what matters to the patient, guided weighing of the pros and cons, and selection of an alternative.[8] When patients consider shared decision-making to have occurred, there is evidence of improved knowledge, attitudinal and emotional outcomes, such as reduced decisional regret, and improved satisfaction.[9]

## Preparing people for kidney failure

International guidelines recommend that people at high risk of kidney failure plan their future care.[10] Key to understanding why shared decision-making and preference-sensitive care have become axioms in this setting is the lack of evidence that survival is sufficiently better with any dialysis modality for it to be considered *best*.[11] Deciding to embark on preparation for a kidney failure treatment represents a weighty decision, firmly embedded within individuals' life and values.[12] People with advanced kidney disease prefer involvement in their health care decisions. Only 8% of US patients with advanced chronic kidney disease favor a passive approach to decision-making.[13] Mismatch between preferred and experienced levels of decision-making engagement is associated with treatment dissatisfaction.[14]

Analysis of nearly 100 interviews with transplant and dialysis recipients[15] identified positive and negative characteristics of kidney failure treatments. Positive characteristics included freedom (e.g., to carry out daily activities and travel), convenience (including proximity to home), independent self-care, treatment effectiveness, simplicity, social inclusion (including camaraderie with patients and staff), and security (including reassurance from clinicians). Negative characteristics included confinement limiting activities and travel, perceived risk of treatments, family burdens, pain, time commitment, subordination to hospital rules, impacts of dialysis access, self-care responsibilities, the transient nature of therapies, and the need for home modifications. Multiple other studies corroborate that independence, safety, quality and quantity of life, and flexibility in daily schedule are central to people's decisions.[16] *Survival* for people with advanced kidney disease does not simply refer to years lived, but to the capacity to cope with the challenges posed by a life with kidney failure.[17] This means that many may be more concerned about the quality of their lives than with the length.[16] Table 1 highlights research demonstrating how patients' preferences for kidney failure treatments underlie decisions about individual modalities and span treatment approaches.

## Dialysis

Those considering dialysis prioritize convenience, flexibility, ability to travel, dialysis-free days, ability to work and socialize, hours on dialysis, better health, and staff support.[15,20] People consider their prospects of transplantation, with dialysis often perceived as an interim measure or second-best alternative.[20] Except for occasional references to company, camaraderie, and clinical contact, dialysis attributes tend to be presented as negatives. The complex and onerous routine centered around treatment appears crucial.[20]

**Table 1 | Selected publications highlighting patient preferences across kidney failure therapies, and how these underlie decisions regarding individual treatments**

| Evidence | Dialysis | Conservative kidney management | Kidney transplantation |
|---|---|---|---|
| Spanning treatment approaches | De Jong et al. 2021[18]: *survey of 7820 patients receiving kidney replacement therapy across 38 countries*. Key drivers influencing treatment choice included quality of life, survival, and safety. Priorities appeared similar across countries with diverse economies.<br>Urquhart-Secord *et al.* 2016[17]: *nominal group technique to identify and rank priorities with 58 patients and 24 caregiver Australian participants*. Patients prioritized outcomes relevant to daily living and well-being above clinical outcomes such as death, adverse events, and biological markers. *Survival* was interpreted by participants as the capacity to cope with the challenges posed by a life with kidney failure.<br>Morton *et al.* 2010[19] and de Jong 2021[16]: *syntheses of qualitative studies.* Patient preferences included pursuit of normality and control and did not match those of health care professionals. Information timing and quality inhibited choice of home treatments and conservative management. | | |
| Specific to treatment approach | Morton *et al.* 2010[15]: *qualitative interviews with 95 Australian patients receiving kidney replacement therapy* and 2011[20]: *nominal group technique with 34 Australian* predialysis *patients and caregivers*. The most important characteristics of dialysis were survival, convenience of dialysis at home, and dialysis-free days.<br>Harwood and Clark 2013[12]: *synthesis of qualitative studies*. Minimizing intrusion from dialysis was the theme that most influenced decisions between modalities. Individual approaches to pursuit of *normality* manifested as favoring in-center *or* home treatment.<br>Walker *et al.* 2018[21]: *discrete choice experiment with 143 New Zealand adults with advanced kidney disease*. Patients preferred home over in-center dialysis when increased nursing support was available and when longer survival, well-being, and flexibility were expected. | Hole *et al.* 2025[22]: *qualitative interviews with 15 older English patients with advanced chronic kidney disease.* Those who chose conservative kidney management considered the negative aspects of dialysis to outweigh the possible survival benefits.<br>Morton *et al.* 2012[23]: *discrete choice experiment with 105 Australian patients with chronic kidney disease.* Patients were more likely to choose dialysis where this provided better life expectancy. They were more likely to choose conservative kidney management for fewer visits to hospital and less-restricted travel.<br>Hole *et al.* 2025[24]: *discrete choice experiment with 327 older English adults with advanced chronic kidney disease.* Participants needed 8%–59% absolute survival benefit 2 years after starting treatment to accept dialysis, with preferences for less-frequent treatment and treatment at home. | Jones *et al.* 2023[25]: *qualitative evidence synthesis including 37 studies*. Freedom from the constraints and demands of dialysis motivated preferences for transplantation. Meanwhile, people had complex and diverse preferences around choice of donor and the balance of risks and benefits from transplantation.<br>Howell *et al.* 2017[26]: *best-worst choice experiment with 93 Australian transplant recipients*. Transplant recipients regarded graft loss as worse than death and showed minimal willingness to trade graft loss for an improvement in any other outcome.<br>Mehrotra *et al.*[27] 2022: *discrete choice experiment with 605 American people waiting for or having received a transplant*. Participants preferred accepting a lower-quality kidney in return for shorter waiting time, particularly those who were older and had lower functional status. |

Some treatment characteristics can be pros for some and cons for others,[15] for example, the opportunity to self-care versus the burden of doing so.[12] Individuals' preferences for clinical versus self-administered treatment, presence of dialysis equipment in the home, self-needling, and abdominal catheters can therefore act as reasons for or against home dialysis.[19,28] Socioeconomic factors, including home environments and the implications of travel for in-center treatment, influence choice of home therapies.[21] Reducing disruption of family and cultural activities, work, social life, and finances can be more important than the survival implications of treatments,[12,28] particularly for Indigenous peoples.[29]

Multiple discrete choice experiments have shown that some people would opt for the potential benefits of longer or more frequent dialysis sessions. Two-thirds of in-center hemodialysis recipients would consider longer treatment sessions if these provided better survival or quality of life, reduced the need for fluid restriction, or prevented fistula problems.[30] Patients appear more likely to choose frequent home treatments that offer better survival.[21] However, in one analysis, 44% would not choose daily hemodialysis regardless of the benefits.[31] Those who would switch required increased life expectancy and quality of life, reduced hospitalization, or less transport time.

## Transplant

Kidney transplantation improves survival and quality of life, while providing freedom from dialysis.[32] People with kidney failure value transplantation and graft function highly.[25] Patients appear willing to accept substantial risks of cancer, dying, or cardiovascular disease to maintain graft survival.[26] Meanwhile, people are highly averse to transplant waiting time. A discrete choice experiment explored the trade-offs patients would make when deciding between accepting an available kidney and waiting for a future offer.[27] Respondents would accept a kidney offering 6.5 years of graft survival, rather than wait 2 years for one expected to last 11 years.

However, for some people, the potential negatives of kidney transplant—death; immunosuppression; cultural, spiritual, and existential concerns; uncertainty; and risk—can outweigh the possible benefits.[25] Individuals may postpone or decline a transplantation opportunity due to personal reasons, such as financial or educational commitments.[25] Some, especially older people, may prefer that a donated kidney go to a recipient who would benefit more from it.[25]

### Conservative kidney management and end-of-life care

For many people, the desire to live longer is the driving force behind initiating dialysis. Reflecting current observational evidence that older, comorbid people do not consistently experience better survival with dialysis,[33] conservative kidney management is increasingly seen as a potentially beneficial alternative. Meanwhile, those who opt for conservative kidney management do not do so simply because the survival benefit fades, but because for them the benefits of dialysis do not outweigh the burdens.[22] The preferences behind these decisions are the same as those already described. Transport, intrusion into the home, and life *tied to a machine* are important negatives that favor conservative kidney management.[19] Others include dialysis access, fatigue, frustration, boredom, time *lost* to treatment, and potential dialysis complications.[34,35]

Choice experiments show that people will trade off considerable life expectancy to reduce the burdens, costs, and restrictions of dialysis. Evidence from Singapore suggests that making people aware of the option of conservative kidney management may lead to more people choosing it, even where the survival outcomes were less than those from dialysis.[36] A study in Australia found that patients would trade off 7 months of life expectancy to reduce the number of hospital visits by 1 per week and 15 months to lessen their travel restrictions.[23] UK patients will trade off approximately 10% absolute survival 2 years after treatment initiation to reduce the frequency of hospital dialysis from 5 to 3 times per week.[24]

### Potential deficiencies in preference-sensitive kidney care

The importance of involving patients and their families in kidney failure treatment decisions has been recognized for decades, but inadequacies remain widespread.[13,14,37] Patients often perceive that they had insufficient information, decision-making control, and support.[19] Many appear to make decisions with incomplete knowledge of their options and inadequate understanding of the likely outcomes.[19] Patients' choices might be resisted, and they can encounter power dynamics with clinicians and fragmented care and advice from disparate teams.[38] Almost half of hemodialysis recipients thought that the choice to start their modality was not their own.[14,37] The low rates of home therapies seen in many settings likely reflect a failure to ensure that patients' preferences guide treatment choices. Unwarranted variation in uptake of kidney replacement therapy and conservative kidney management suggests the same for decisions between these treatments.[39]

Deficiencies may reflect lack of clinician enthusiasm for and competence in preference-sensitive care.[40] How clinicians communicate with patients and families regarding treatment options is highly variable.[19] Overall, clinicians tend to promote dialysis and encourage initiation,[41,42] even when they do not consider dialysis to be the best treatment. The information documents to which patients are routinely exposed are often similarly biased, framing dialysis as having minimal side effects or complications and conveying unrealistic expectations about life expectancy and quality of life.[43] Tensions can arise when clinicians' and patients' priorities differ,[25,38] and clinicians can resist and question patients' goals and preferences.[44] Rather than embracing a person-centered approach, clinical guidelines define the value of treatments based on their ability to extend life or to improve surrogate markers associated with survival, such as small molecule clearance. Unsurprisingly, people living with advanced kidney disease perceive an immense pressure to start dialysis,[38] while they would often prefer to postpone or avoid the upheaval and constraints of treatment planning and initiation.[45,46]

### Discussion and recommendations

The evidence summarized in this review suggests a mismatch between contemporary kidney care and the person-centered approach needed to ensure that patients' preferences guide when and which kidney failure treatments are initiated. Patients' preferences for kidney failure treatments appear remarkably transferrable between settings, populations, and studied approaches. People planning for life with kidney failure trade off the potential benefits of treatments against an over-riding desire to minimize disruption of daily and family life, responsibilities, and interests. The fact that such trade-offs motivate their actions explains why people will accept burdensome and intrusive kidney replacement therapies, opt for conservative kidney management, postpone dialysis initiation, discontinue dialysis, omit dialysis sessions, and decline transplantation. Figure 1 attempts to capture the complex processes and multiple actors involved in kidney failure treatment decision-making. Knowledge of what is important to patients and caregivers is essential to optimize care. Table 2 contains recommendations as to how clinicians and services can reframe the culture and purpose of kidney care to be more about person-centeredness.

Efforts have been made to advance preference-sensitive kidney care. Decision support tools are available and can increase knowledge of treatments and outcomes, facilitating preference sensitivity.[50] However, quality and content are mixed, and they are not widely used.[48] The Standardized Outcomes in Nephrology (SONG) initiative[51] reframes treatment outcomes based on the shared priorities of patients, caregivers, and other relevant stakeholders. The International Consortium for Health Outcomes Measurement (ICHOM) initiative[52] developed a standardized set of

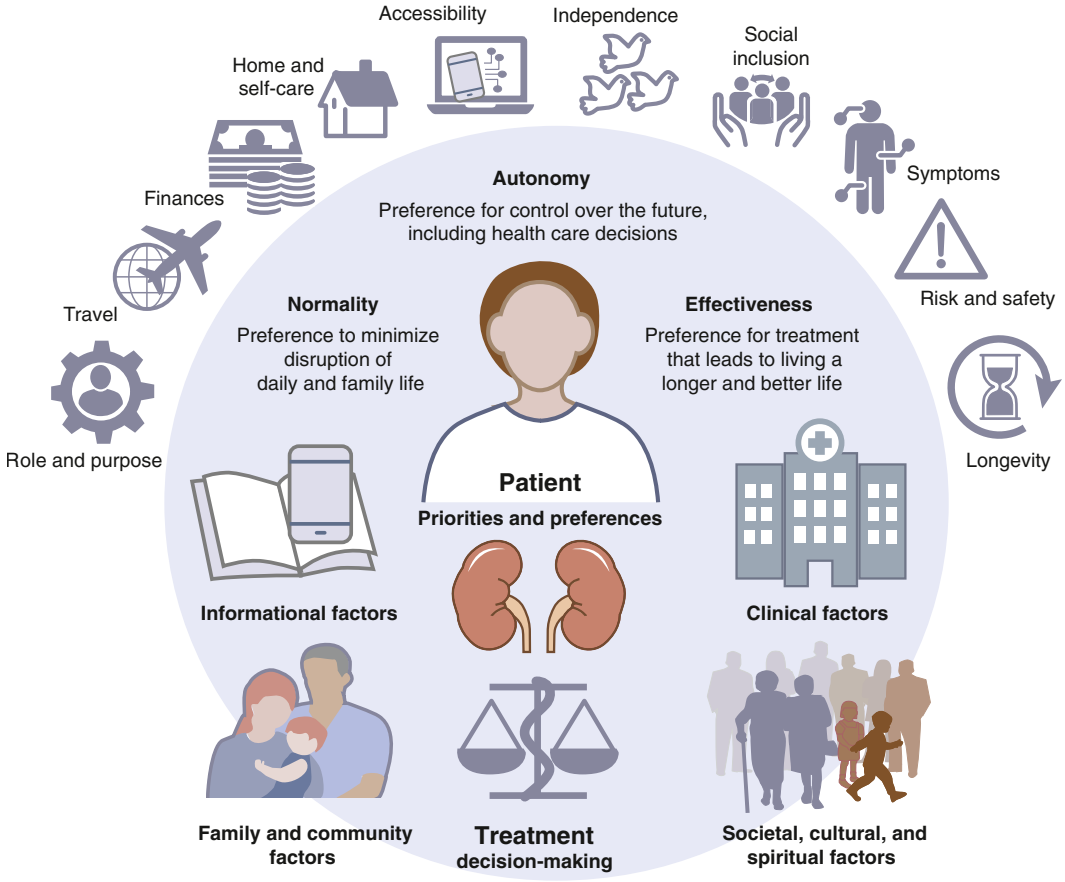



**Figure 1 | Patients have diverse preferences that reflect individual variation in what they consider important**. Those planning for life with kidney failure want control over their futures and tend to desire autonomy in treatment decision-making. They will trade off the potential benefits of treatments against an over-riding desire to minimize disruption of daily and family life, responsibilities, and interests. The treatments that patients plan for and receive reflect complex processes involving many actors. Familial, community, societal, cultural, spiritual, clinical, and informational factors shape preferences, restrict or provide treatment options, and guide decisions. Skill, investment, and intentional practice are needed to ensure that patient preferences carry weight in decision-making.

patient-centered outcomes for clinical use. However, these initiatives will succeed only if accompanied by a shift in the paradigm of what it means to *do nephrology* well. Critically, those involved in shaping and delivering care must recognize that living longer is not preeminent for people living with kidney failure.

There are limitations in our knowledge of patient preferences for kidney care. Despite initiatives such as SONG,[51] much research fails to capture patient-important outcomes. The evidence available is dominated by studies in Australia, Northern Europe, and the USA. How preferences differ between, and whether person-centered kidney care must be adapted for, people with different cultural backgrounds requires consideration. We also need discussion at a societal and political level: to what extent, and in what circumstances, should clinicians deliver care that accords with patient preferences but goes against their own clinical judgment of how quantity and quality of life should be optimized? Health care providers must identify therapies that are likely to be successful and offer them as options to patients. Comprehensive shared decision-making includes patients' preferences in this step, because *success* looks different to different people, depending on their goals and preferences. We must accept the notion that patients might make decisions that we ourselves would not choose. As long as a treatment's outcomes might represent *success* to a patient, such decisions must be supported to achieve the goal of person-centered care. Meanwhile, it is unethical to use individual patient preferences as an excuse to abdicate responsibility for harm—or indeed to appeal to the principle of autonomy and abandon patients to make decisions on their own. Clinicians must support patients to understand the potential benefits and harms of treatment choices and help them align their preferences with realistic expectations of treatment outcomes.

Although this mini review spotlights preference sensitivity in decisions between kidney failure treatments, person-centered kidney care must not be reduced to allocation of people to the *best* treatment. A diagnosis of advanced kidney disease can be intensely upsetting and carries profound implications for autonomy, normality, and social identity.[17] Patients' preferences for control, independence, achievement

**Table 2 | Biomedical and person-centered paradigms of nephrology practice, with recommendations at clinician and service/profession level to facilitate preference-sensitive care**

| Biomedical vs. person-centered paradigm | Clinician-level recommendations | Service/profession level recommendations |
|---|---|---|
| *The best treatment for kidney failure is that which most effectively controls pathology, manages symptoms, and prolongs life.*<br><br>vs.<br><br>The best treatment for kidney failure is the one most aligned with the recipient's values, goals, and preferences. Informed, well-supported patients will sometimes decide to decline, discontinue, or disengage from biomedically *effective* treatments, including dialysis and transplantation. | Clinicians must develop ways to ask their patients what is important to them. Leading with preferences and values enables patients to apply what is important to them during treatment decision-making.[41]<br><br>Suggested approaches:<br>• Ask patients *What is most important to you?*<br>• Terms such as *predialysis* care should be avoided.<br>• Use tools such as IPOS-renal[47] to ascertain patients' priorities.<br>• Describe the availability of home and in-center dialysis, conservative kidney management, and transplantation to *all* patients.<br>• Do not persuade patients to accept treatments that are not aligned with their preferences. | Embrace person-centered care as an institutional philosophy. Ensure that time, space, and treatment flexibility permit individuals' preferences to guide care.<br><br>Suggested approaches:<br>• Leaders in kidney medicine should work with patient representatives to promote and role-model preference-sensitive care.<br>• Clinicians should be instructed in person-centered care during undergraduate and postgraduate education, including communication skills training.<br>• Clinical studies must collect outcomes of importance to patients. |
| *Clinicians can identify the* right *treatment for a clinical problem.*<br><br>vs.<br><br>Patients hold values and preferences, which define the right treatment. Even when only one treatment is available, the right course of action for some patients may be to say *no*. | Clinicians must not abuse the trust and power afforded by their position. Unquestioning implementation of guidelines, however strong the evidence base, should be avoided.<br>Suggested approaches:<br>• Highlight that planning for the possibility of kidney failure involves patient choice.[41]<br>• Ask patients to invite friends and family to help them with decision-making.<br>• Share wlhat is known about how effective kidney failure treatments are at prolonging life and maintaining independence along with their burdens.<br>• Help people to weigh these benefits and burdens based on what is acceptable to them and what they want to achieve.<br>• Use patient decision aids to support decisions.[48] | Those defining *quality* kidney care must consider the full effects of guidance, including perverse incentives. Services must embed shared decision-making—the cornerstone of person-centered care.<br>Suggested approaches:<br>• Informational materials should be high quality and unbiased.[43]<br>• Person-centered decisions, rather than guideline-compliant care, should be remunerated.<br>• Guidelines should be explicit when prioritization of life prolongation is implicit—so that patients and clinicians understand the reasoning behind recommendations.<br>• Guidelines should clearly convey the quality of evidence, enabling clinicians and patients to consider its applicability. |
| *Those with advanced chronic kidney disease should make smooth, healthy transitions into kidney failure.*<br><br>vs.<br><br>Individuals should be supported to live as well as possible, despite the possibility of future kidney failure. Although preparing for a treatment modality is goal-concordant for many patients, others prefer to postpone or avoid the upheaval and constraints of treatment planning and initiation. | Skillful, individualized care is needed to support people to decide how best to live with the possibility of developing kidney failure. Clinicians must take care not to reinforce a perspective that dialysis is best or recommended, without first establishing the person's preferences.<br>Suggested approaches:<br>• Describe the *reasons* rather than the *need* for preparation and initiation of kidney replacement therapy.[41]<br>• Avoid reinforcing a *do dialysis or die* concept, which strips patients of the perception of choice.[22]<br>• Recognize that some patients will *decide not to decide*.[49]<br>• In the absence of life-threatening complications of low kidney function, use supportive and medicinal approaches to symptom control before considering dialysis initiation. | Service aims must be reconciled with the person-centered approach. Barriers to the uptake of therapies that aligned with patients' preferences for normality and reduced treatment intrusion should be identified and addressed.<br><br>Suggested approaches:<br>• Use the uptake of home therapies and conservative kidney management as metrics of success in delivering person-centered care.<br>• Offer extra help to those who started (hemo)dialysis without time to prepare for their chosen modality.<br>• Availability of individualized dialysis prescriptions (such as incremental regimens) may allow patients to express their preferences for these.<br>• Provide multidisciplinary expertise in management of advanced chronic kidney disease, which may enhance uptake of conservative kidney management and postpone dialysis initiation. |

IPOS, Integrated Palliative Outcome Score.

of life goals, and support for loved ones reflect general care priorities of the seriously ill.[53] Patients whose wider needs are met are more likely to be able to plan well for a future living with kidney failure. Evidence-based approaches are needed to ensure comprehensive, preference-sensitive care, including economic analyses that focus on quality of life associated with treatment choices; research and practice from palliative care may be relevant here.

In conclusion, this review highlights how people's preferences can guide decisions between kidney failure treatments, but only if clinicians and systems facilitate such decision-making. The best treatment for a person is the one that fits best with what is important to them. The *science* of nephrology involves optimal delivery of treatments that prolong life, and preserve and replace organ function. The *art* lies in shaping this care to the person facing kidney failure.

## DISCLOSURE

All the authors declared no competing interests.

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